Uniform Slater-orbit densities for linear selected preparations

Exact energy pinching in the one-excitation sector

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Abstract

We prove a finite-dimensional, size-independent moment bound for the orbit density of a linear Slater-adapted preparation. The key estimate controls both variance operators of a rectangular Gaussian synthesis map. Its input covariance is bounded using the Weyl orbit of the Slater weight, with representation multiplicities retained. Standard matrix-Gaussian integration by parts then bounds every fixed density moment. For the normalized selected creation operator the second moment is at most three, uniformly in the microscopic parameters. This transfers exact energy-pinching errors from the covariant trace to the one-selected excited state and closes its actual neutral-slope correlators. Nonlinear selected preparations are left for a separate estimate.

1 Finite-N setting and the result

Let $\mathcal{H}_m = \bigwedge^m \mathbb{C}^N$, $d_m = \binom{N}{m}$ and $|O\rangle$ be a fixed Slater determinant. Use the original overlapping complex Hermitian Gaussian p-body couplings. Express their GUE law in real independent standard Gaussian coordinates g_a. Physical $U(N)$ rotations act on these coordinates by an orthogonal representation $R(U)$ and on $\mathcal{H}_m$ by $U_m = \bigwedge^m U$.

Take a homogeneous linear preparation $P_O(J)$ satisfying

$$P_O(k \cdot J) = k_m P_O(J) k_m^\dagger, \qquad k \in U(m) \times U(N-m), \quad a_N = \mathbb{E}\,\|P_O(J)|O\rangle\|^2 . \tag{1}$$

The subgroup preserves the occupied plane. Normalized $M_N^\dagger$, h_N, u_N and fixed linear combinations obey this condition. For $M_N^\dagger$, $a_N = 1$ exactly.

Theorem 1. *There is a positive covariant random matrix $\mathcal{R}_{P,N}$ such that, for every covariant $Y_N(J)$ with $\mathbb{E}\,\mathrm{tr}_m(Y_N^\dagger Y_N) < \infty$,*

$$\mathbb{E}\langle O|P_O^\dagger Y_N P_O|O\rangle = \mathbb{E}\,\mathrm{tr}_m(\mathcal{R}_{P,N} Y_N), \qquad \mathbb{E}\,\mathrm{tr}_m\,\mathcal{R}_{P,N}^r \le (2r-1)!!\, a_N^r \tag{2}$$

for every integer $r \ge 1$. Here $\mathrm{tr}_m = \mathrm{Tr}\,/d_m$.

This theorem is finite N; it does not assume double scaling or replace finite-N spectral projectors by their limiting functions.

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2 The orbit synthesis map and its output covariance

Haar measure on $G = U(N)$ is normalized to one. Put

$$q_U(J) = U_m^\dagger P_O(U \cdot J)|O\rangle, \qquad Z(J)f = \sqrt{d_m} \int_G q_U(J) f(U)\, \mathrm{d}U. \tag{3}$$

The map $Z : L^2(G) \to \mathcal{H}_m$ is a finite Gaussian matrix series after restricting its domain to the finite-dimensional matrix-coefficient space that supports all its entries. Define

$$\mathcal{R}_{P,N}(J) = Z(J)Z(J)^\dagger = d_m \int_G |q_U(J)\rangle\langle q_U(J)|\, \mathrm{d}U. \tag{4}$$

Changing the Haar variable proves covariance of this matrix. Rotating the Gaussian variable and using $Y(U \cdot J) = U_m Y(J) U_m^\dagger$ proves the exact pairing identity in Theorem 1.

For the output covariance, Gaussian invariance first removes the rotation from $P_O(U \cdot J)$ under expectation. Schur averaging in the irreducible exterior-power representation then gives

$$\mathbb{E} Z Z^\dagger = d_m \int_G U_m^\dagger \mathbb{E}[P_O(J)|O\rangle\langle O|P_O(J)^\dagger] U_m \, \mathrm{d}U = a_N I_m. \tag{5}$$

Controlling only (5) would not suffice: a Gaussian column $g \in \mathbb{C}^d$ has $\mathbb{E} g g^\dagger = I$ but $\mathbb{E}\,\mathrm{tr}(gg^\dagger)^2 = d+1$. The input variance must also be bounded.

Write $q_U = \sum_a g_a f_a(U)$ and introduce its coefficient vector

$$\Psi_U = \sum_a e_a \otimes f_a(U) = [R(U)^T \otimes U_m^\dagger]\Psi = \rho(U^{-1})\Psi, \qquad \rho = R \otimes \bigwedge^m, \quad \|\Psi\|^2 = a_N. \tag{6}$$

The real Hermitian noise basis makes $R(U)^T = R(U^{-1})$. In ordered complex J_{AB} labels the equivalent noise factor is the conjugate adjoint action, so its torus weight is $B - A$; the physical hop state has weight $O + A - B$.

For a diagonal unitary D, equation (1) gives $q_D = \chi_O(D)^* q_I$, where $\chi_O(D) = \prod_{i \in O} D_{ii}$. Consequently

$$\rho(D)\Psi = \chi_O(D)\Psi. \tag{7}$$

The noise and physical weights therefore sum to the Slater weight. This exact selection rule is what controls the other variance.

3 The Slater weight bounds the input covariance

Decompose the finite-dimensional unitary representation as $\bigoplus_\alpha V_\alpha \otimes \mathcal{M}_\alpha$ and let Ψ_α be the component of Ψ. Equation (7) holds in each nonzero component. The corresponding irreducible representation V_α therefore contains the weight $(1^m, 0^{N-m})$.

Permutation matrices take that weight to every choice of m occupied coordinates. These d_m weights are distinct, and their weight spaces are mutually orthogonal. Hence

$$d_\alpha := \dim V_\alpha \geq d_m \qquad \text{whenever } \Psi_\alpha \neq 0. \tag{8}$$

No highest-weight assumption or multiplicity-free decomposition is needed. Schur averaging, with multiplicity spaces retained, gives

$$F := d_m \int_G |\Psi_U\rangle\langle\Psi_U| \, \mathrm{d}U \tag{9}$$

$$= \bigoplus_\alpha \frac{d_m}{d_\alpha} I_{V_\alpha} \otimes \sigma_\alpha, \qquad \sigma_\alpha = \mathrm{Tr}_{V_\alpha} |\Psi_\alpha\rangle\langle\Psi_\alpha|. \tag{10}$$

Since $\sigma_\alpha \geq 0$ and $\|\sigma_\alpha\| \leq \mathrm{Tr}\, \sigma_\alpha = \|\Psi_\alpha\|^2 \leq a_N$, it follows that $0 \leq F \leq a_N I$.

Let $Af = \sqrt{d_m} \int_G \Psi_U f(U) \mathrm{d}U$. Then $AA^\dagger = F$, while the kernel of $A^\dagger A$ is $d_m \langle \Psi_U, \Psi_V \rangle$. Gaussian covariance shows that this is precisely the kernel of $\mathbb{E} Z^\dagger Z$. The nonzero spectra of $AA^\dagger$ and $A^\dagger A$ coincide, so

$$0 \leq \mathbb{E} Z^\dagger Z \leq a_N I, \qquad \mathrm{Tr}(\mathbb{E} Z^\dagger Z) = d_m a_N. \tag{11}$$

The bounds (5) and (11) are independent of N, p, m except through the preparation norm a_N.

For the selected creation block, this norm is one: it creates the $L_N = \binom{m}{p}\binom{N-m}{p}$ orthogonal p-hop states from O, with covariance $1/L_N$ per label. This also gives an elementary check on (5): the unrotated output covariance is the p-excitation projector divided by L_N, whose Haar average is I_m/d_m.

4 Density moments and transfer of small covariant errors

Hermitize the rectangular series:

$$S = \begin{pmatrix} 0 & Z \\ Z^\dagger & 0 \end{pmatrix} = \sum_a g_a A_a, \qquad V = \mathbb{E}S^2 = \begin{pmatrix} a_N I_m & 0 \\ 0 & \mathbb{E}Z^\dagger Z \end{pmatrix}. \tag{12}$$

We have $V \le a_N I$ and $\operatorname{Tr} V = 2d_m a_N$. For $r \ge 2$, the standard Gaussian integration-by-parts identity [1, Sec. 6] and the derivative of a matrix power give

$$\mathbb{E}\operatorname{Tr} S^{2r} = \sum_a \sum_{k=0}^{2r-2} \mathbb{E}\operatorname{Tr}(A_a S^k A_a S^{2r-2-k}). \tag{13}$$

In the eigenbasis of S, each summand is $\sum_{ij} |(A_a)_{ij}|^2 x_j^k x_i^{2r-2-k}$. Taking absolute values and applying weighted arithmetic–geometric mean bounds it by $\operatorname{Tr}(A_a^2 S^{2r-2})$. This is the usual matrix-Gaussian moment argument [1, Sec. 7]. Thus

$$\mathbb{E}\operatorname{Tr} S^{2r} \le (2r-1)\mathbb{E}\operatorname{Tr}(V S^{2r-2}) \le (2r-1)a_N \mathbb{E}\operatorname{Tr} S^{2r-2}. \tag{14}$$

Start at $\mathbb{E}\operatorname{Tr} S^2 = \operatorname{Tr} V$, rather than at the ambient dimension. Since $\operatorname{Tr} S^{2r} = 2\operatorname{Tr}(ZZ^\dagger)^r$, induction proves

$$\mathbb{E}\operatorname{tr}_m \mathcal{R}_{P,N}^r \le (2r-1)!! a_N^r. \tag{15}$$

If $a_N = 0$, the statement is immediate. The general Gaussian moment method is established literature; the new input here is the orbit variance estimate needed to apply it uniformly.

Let $\|X\|_{r,m} = (\mathbb{E}\operatorname{tr}_m |X|^r)^{1/r}$. Hölder on the joint disorder/matrix trace space now gives

$$|\mathbb{E}\langle O|P_O^\dagger Y P_O|O\rangle| \le \sqrt{3}\, a_N \|Y\|_{2,m}, \tag{16}$$

$$\mathbb{E}\,\|XP_O|O\rangle\|^2 \le \sqrt{3}\, a_N \|X\|_{4,m}^2 \tag{17}$$

for covariant Y, X, respectively. Positivity of Y is not required in the first bound.

For a covariant post-preparation $B(J)$ there is also the exact identity $\mathcal{R}_{BP} = B\mathcal{R}_P B^\dagger$. Consequently

$$\|\mathcal{R}_{BP}\|_{r,m} \le \|B\|_{4r,m}^2 \|\mathcal{R}_P\|_{2r,m}. \tag{18}$$

Fixed covariant polynomials have the required high moments; actual unitary phases preserve them. The preparation order is $BP|O\rangle$. It has not been interchanged with $PB|O\rangle$.

5 The actual neutral slope with exact energy pinching

Return to half filling and the double-scaling limit $\lambda_N = p^2/N \to \lambda > 0$. In the notation of the preceding stage, $h_N = H_N/\sqrt{C_m}$ and

$$t_N = \frac{m\epsilon\alpha[\Phi_m(H_N) - \bar{\kappa}_m H_N]}{8\lambda_N C_m^{3/2} q_N \sqrt{\lambda_N}}, \qquad b_N = (1 - \mathcal{P}_{H_N}) t_N. \tag{19}$$

This b_N is the normalized actual neutral slope. The pinching $\mathcal{P}_{H_N}$ preserves whole degenerate energy blocks. Time is dimensionless, conjugate to h_N; physical string energy is $(J_s/\sqrt{\lambda})h$.

The preceding central-sector argument already proved

$$X_N := \mathcal{P}_{H_N} t_N, \qquad \|X_N\|_{r,m} \longrightarrow 0 \quad \text{for every fixed finite } r. \tag{20}$$

It uses positivity of the two-energy measure, zero diagonal mass in its q-Mehler limit, Portmanteau, high moments and pinching contractivity. It does not interchange infinite time with N. Applying (16) with $P = M_N^\dagger$, $a_N = 1$, gives

$$\mathbb{E}\left\| X_N M_N^\dagger |O\rangle \right\|^2 \le \sqrt{3}\, \|X_N\|_{4,m}^2 \longrightarrow 0. \tag{21}$$

This is the missing finite-N estimate for the one-selected state.

For any fixed word W of covariant insertions, telescope $W(h_N, b_N) - W(h_N, t_N)$ and use tracial Hölder with sufficiently high fixed moments. Its S_2 norm tends to zero. The first bound in (16) transfers that difference to the selected state. The already proved unpinched same-J limit then yields

$$\lim_N \varphi_N\big[M_N W(h_N, b_N) M_N^\dagger\big] = \tau\big[\rho_1 W(h, u)\big], \tag{22}$$

where τ is the standard two-color q-Gaussian trace and

$$\rho_1 = (\gamma h - \sqrt{\lambda}\gamma u)^2 + [1 - \sqrt{q}(1 + \lambda)]I, \qquad \gamma = q^{1/4}. \tag{23}$$

Any finite number of slope insertions and fixed actual h_N time phases inside the preparation sandwich are allowed. In particular,

$$\lim_N \varphi_N(M_N b_N^2 M_N^\dagger) = 1 + \lambda\sqrt{q}(1 + q). \tag{24}$$

The finite-N operation here is exact pinching, followed by $N \to \infty$. It is stronger than the previous ordered finite-window statement for this preparation class.

6 Exact finite checks, scope and the next interface

Five finite Slater fixtures check 59 noise-plus-physical weight identities and 122 permutation-frame counts. Two exact stabilizer matrices for each of $(N, m, p) = (4, 2, 1), (4, 2, 2)$ check 624 entries, including conjugation of the noise action and CAR signs.

An independent projective case provides the complete density. For $m = p = 1$, let J be the single-particle Hermitian GUE matrix. Since $q_\psi = (I - |\psi\rangle\langle\psi|)J|\psi\rangle/\sqrt{N-1}$, Haar integration gives

$$\mathcal{R}(J) = \frac{NJ^2 - 2\,\mathrm{Tr}(J)J + [(\mathrm{Tr}\,J)^2 + \mathrm{Tr}\,J^2]I/(N+1)}{(N-1)(N+2)}. \tag{25}$$

The checker independently expands the three projector factors using

$$\mathbb{E}_\psi \prod_{\ell=1}^{k} (|\psi\rangle\langle\psi|)_{i_\ell j_\ell} = \frac{\sum_{\pi\in S_k} \prod_\ell \delta_{i_\ell, j_{\pi(\ell)}}}{N(N+1)\cdots(N+k-1)}, \qquad k \le 3. \tag{26}$$

This formula follows by twirling onto the symmetric tensor power. Two exact complex fixtures at each $N = 2, 3, 4$ verify all 58 entries of (25) and its principal-minor positivity. Only $N = 2$ is half filled in this diagnostic family.

Independent Gaussian-coordinate Wick contraction at $N = 2, 3, 4, 5$ checks the mean I and second moment with 978 coordinate pairs. Writing $D = (N-1)(N+2)$, the result is

$$\mathbb{E}\,\mathrm{tr}\,\mathcal{R}^2 = \frac{2N^4 - 5N^2 + 3 + \frac{2N+3}{(N+1)^2}(N^4-1)}{D^2} \le 3. \tag{27}$$

At $N = 2$, $\mathcal{R} = (\chi_3^2/3)I$; its exact moments $\prod_{j=0}^{r-1}(3+2j)/3^r$ obey (15). Orders one through eight are saved. No Haar Monte Carlo or asymptotic fit is used.

These checks supplement the general proof. The first checker attempt compared differently simplified complex expressions; entrywise exact simplification resolved the failed assertion. The initial diagnostic is retained. Successful scientific output is sealed, with source, versions, PDF build and renders bound by a separate SHA256 manifest.

The result covers linear preparations and covariant post-dressing, including all covariant correlators in the $M_N^\dagger|O\rangle$ state. For multiple interlaced selected insertions, or a preparation $M_N^\dagger B_N|O\rangle$, the synthesis map is generally nonlinear Gaussian and the present moment recurrence does not apply. The next task is that larger preparation class and the J_2/complete bilocal interface, followed by the physical spatial, mass and finite-G dictionary.

References

[1] J. A. Tropp, *Second-Order Matrix Concentration Inequalities*, arXiv:1504.05919; author's preprint, Secs. 6–7, Lemma 6.1 and Propositions 7.1, 7.3.